



SENDER TECHNICAL INTEGRATION GUIDE

API Specifications • Security • Workflow • JSON Schema

DOCUMENT PARTICULARS

Receiver:	Alshumookh Global Banking Financing And Credit
Platform:	https://api.alshumookh-pay.com
Issued By:	ALSHUMOOKH GROUP — Technology & Compliance
Purpose:	API-to-API Technical Integration Reference
Date:	May 2026 STRICTLY CONFIDENTIAL

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RECEIVER PROFILE

Entity Name	Alshumookh Global Banking Financing And Credit
Role	Receiver / Execution Party — Digital Asset Transactions
Jurisdiction	Dubai, United Arab Emirates
Office Address	API World Tower, Office No. 2103/2104, Shaikh Zayed Road, Dubai, UAE
License No.	887065
Authorised By	Ahmed Mohamed — Chief Executive Officer
Contact Email	ceo@alshumookhgroup.ae
Platform URL	https://api.alshumookh-pay.com
Ingest Endpoint	POST https://api.alshumookh-pay.com/api/v1/payloads/ingest
OAuth2 Token	POST https://api.alshumookh-pay.com/api/v1/oauth/token
Receiver Wallet	0xBD682cfD8382a90adfDd6745780D3D7959c4d939

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SECURITY ARCHITECTURE — ALL ACTIVE CONTROLS

 HMAC-SHA256 Payload Signing	 OAuth 2.0 Client Credentials
 JWT / JWS Crypto Verification	 mTLS Ready Cert Fingerprint
 Signed Webhooks S2S Notifications	 Reconciliation Full Audit Engine
 WAF Shield Intrusion Protection	 TLS 1.3 / HSTS Transport Security

Security Controls — Technical Specification

Control	Header / Protocol	Technical Detail
HMAC-SHA256	X-Signature	HMAC-SHA256(X-Timestamp + '.' + raw_body). Enforced per-client flag.
OAuth 2.0	Authorization: Bearer	client_credentials grant at POST /api/v1/oauth/token.
JWT / JWS	X-JWS-Signature	Detached JWS with RS256/HS256 via PyJWT. Payload hash in claims.
mTLS Cert	X-Client-Cert-Fingerprint	SHA-256 cert fingerprint forwarded by proxy, verified at app layer.
Signed Webhooks	X-ALSHUMOOKH-Signature	All outbound callbacks HMAC-signed. Alchemy / Coinbase / MoonPay inbound verification.
Reconciliation	POST /admin/reconcile	Full audit trail: tx hash, amounts, timestamps, status per payload.
WAF / Rate Limit	Middleware	IP auto-ban, user-agent classification, path-probe detection.
Replay Guard	X-Timestamp	Timestamp validated ± 5 min window. Idempotency-Key enforced.
Transport	TLS 1.2 / 1.3	HSTS max-age=31536000. CSP, X-Frame-Options, X-Content-Type-Options.
IP Whitelist	Middleware	Per-client allowed IP ranges enforced at gateway layer.

Authentication Modes — Configurable Per Client

Level	Mode	Required Headers	Grade
1	API Key Only	X-API-Key + X-Client-ID	Standard
2	API Key + HMAC	X-API-Key + X-Signature + X-Timestamp	Enhanced
3	OAuth2 Bearer	Authorization: Bearer <token>	Professional
4	OAuth2 + HMAC	Authorization: Bearer + X-Signature + X-Timestamp	Institutional
5	mTLS + Key	X-Client-Cert-Fingerprint + X-API-Key	Maximum
6	OAuth2 + JWS	Authorization: Bearer + X-JWS-Signature	Full Crypto

All authentication modes are provisioned per client at the API-key level. The security tier is assigned based on transaction volume and institutional requirements.

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TRANSACTION WORKFLOW — END TO END

Complete process from sender onboarding through blockchain settlement finality. All 12 steps are mandatory and executed in sequence.

01	☐	KYC & ONBOARDING	Sender submits KYC documentation. ALSHUMOOKH Compliance conducts AML/CTF screening, sanctions check, and source-of-funds review. Internal approval granted before any operation.
02	☐	API KEY ISSUANCE	Upon KYC approval, Sender receives API Key + Client ID via secure encrypted channel. System integration is configured pointing to: https://api.alshumookh-pay.com
03	☐	CREDENTIAL SUBMISSION	Sender provides: Master Wallet Address + Private Key (technical verification only) + Smart Contract Address. This authorises Receiver to conduct controlled on-chain verification.

04	☐	ON-CHAIN VERIFICATION	ALSHUMOOKH verifies: token balance, USDT ERC20/TRC20 smart contract integrity, and execution readiness. No execution proceeds until all on-chain checks pass.
05	☐	API PAYLOAD SUBMISSION	Sender POSTs signed JSON payload to: https://api.alshumookh-pay.com/api/v1/payloads/ingest Must include: X-API-Key + X-Timestamp + X-Signature (HMAC) or OAuth2 Bearer token.
06	☐	PLATFORM VALIDATION	System validates: API Key/Bearer token, HMAC signature, sender_reference uniqueness, wallet format, transaction_hash on-chain, timestamp window (±5 min), and JSON schema.
07	☐	PGL PREPARATION	Receiver prepares Payout/Paymaster Wallet List (PGL) — max 5 beneficiary wallets with percentage distribution totalling exactly 100%. Sender confirms in writing.
08	☐	LIVE EXECUTION SESSION	Mandatory live session via Zoom or Google Meet. Final authorisation checkpoint. Both parties present. Master Wallet imported into controlled secure environment.
09	☐	SMART CONTRACT EXECUTION	Tokens transferred: Master Wallet → Bank Broadcast Wallet → Smart Contract. Distribution executed automatically per approved PGL percentages.
10	☐	SIGNED WEBHOOK CALLBACK	HMAC-signed webhook dispatched to Sender's endpoint confirming: execution status, transaction hash, block number, and per-wallet distribution amounts.
11	☐	BLOCKCHAIN CONFIRMATION	On-chain confirmation hash recorded. Transaction receipts retained for compliance audit. Dashboard updated. Reconciliation engine triggered automatically.
12	☐	SETTLEMENT FINALITY	Transaction deemed final and irreversible upon on-chain confirmation. Formal settlement report issued. Records archived per AML/CTF requirements.

SETTLEMENT COMPLETE — Final and irreversible upon on-chain confirmation. HMAC-signed settlement report dispatched to Sender's registered webhook endpoint.

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COMPLETE ACCEPTED JSON PAYLOAD — FULL SCHEMA

Every request to the ingest endpoint must include a JSON body matching the schema below. Fields marked ★ are mandatory. Amounts are decimal (float). All strings are UTF-8.

4.1 Field Reference

Field	Type	★	Description / Accepted Values
sender_reference	string	★	Unique TX ID from Sender. Max 128 chars. Rejected if duplicate.
client_id	string	★	Client ID issued by ALSHUMOOKH at onboarding.
amount	number	★	Decimal amount > 0. Example: 1000000.00
currency	string	★	Fixed value: "USDT"
network	string	★	Must be: "ERC20" or "TRC20"
sender_wallet	string	★	Master wallet. ERC20: 0x... (42 chars). TRC20: T... (34 chars).
receiver_wallet	string	★	ALSHUMOOKH wallet: 0xBD682cfD8382a90adfDd6745780D3D7959c4d939
smart_contract	string	★	Smart contract address used for token execution.
transaction_hash	string	★	On-chain TX hash — must be verifiable on the specified network.
beneficiary_wallets	array	★	Array of objects. Min 1, max 5. All percentages must sum to 100.
■ wallet	string	★	Beneficiary wallet address (inside beneficiary_wallets).
■ percentage	number	★	Distribution %. Integer or decimal. All items must sum to 100.
timestamp	string	★	ISO-8601 UTC. e.g. 2026-05-11T10:00:00Z. Within ±5 minutes.
notes	string	—	Optional. Free text. Max 500 chars.
webhook_url	string	—	Optional HTTPS callback URL for signed status notifications.
idempotency_key	string	—	Recommended. Prevents duplicate processing on retry.

4.2 Complete JSON Example — All Fields

```
{ "sender_reference": "TXN-2026-001", "client_id": "YOUR_CLIENT_ID", "amount": 1000000.00, "currency": "USDT", "network": "ERC20", "sender_wallet": "0xYOUR_MASTER_WALLET_ADDRESS", "receiver_wallet": "0xBD682cfD8382a90adfDd6745780D3D7959c4d939", "smart_contract": "0xSMART_CONTRACT_ADDRESS", "transaction_hash": "0xBLOCKCHAIN_TX_HASH", "beneficiary_wallets": [ { "wallet": "0xWALLET_ADDRESS_1", "percentage": 40 }, { "wallet": "0xWALLET_ADDRESS_2", "percentage": 35 }, { "wallet": "0xWALLET_ADDRESS_3", "percentage": 25 } ], "timestamp": "2026-05-11T10:00:00Z", "notes": "Optional transaction notes", "webhook_url": "https://your-system.com/webhook/callback", "idempotency_key": "IDEM-2026-001" }
```

4.3 Minimal JSON Example — Required Fields Only

```
{ "sender_reference": "TXN-2026-001", "client_id": "YOUR_CLIENT_ID", "amount": 1000000.00, "currency": "USDT", "network": "ERC20", "sender_wallet": "0xYOUR_MASTER_WALLET_ADDRESS", "receiver_wallet": "0xBD682cfD8382a90adfDd6745780D3D7959c4d939", "smart_contract": "0xSMART_CONTRACT_ADDRESS", "transaction_hash": "0xBLOCKCHAIN_TX_HASH", "beneficiary_wallets": [ { "wallet": "0xWALLET_ADDRESS_1", "percentage": 60 }, { "wallet": "0xWALLET_ADDRESS_2", "percentage": 40 } ], "timestamp": "2026-05-11T10:00:00Z" }
```

05

HTTP REQUEST EXAMPLES

5.1 Standard Request — API Key Authentication

```
POST https://api.alshumookh-pay.com/api/v1/payloads/ingest Content-Type: application/json X-API-Key: YOUR_API_KEY X-Client-ID: YOUR_CLIENT_ID X-Timestamp: 2026-05-11T10:00:00Z X-Idempotency-Key: IDEM-2026-001
Body: { "sender_reference": "TXN-2026-001", "amount": 1000000.00, ... }
```

5.2 Institutional Request — OAuth2 Bearer + HMAC-SHA256

```
# STEP 1 – Obtain Bearer Token POST https://api.alshumookh-pay.com/api/v1/oauth/token Content-Type:
application/x-www-form-urlencoded Body: grant_type=client_credentials&client_id=ID&client_secret=SECRET
Response: {"access_token": "eyJ...", "token_type": "bearer", "expires_in": 3600} # STEP 2 – Compute HMAC
Signature SIGNATURE = HMAC-SHA256( X-Timestamp + "." + raw_json_body, hmac_secret ) # STEP 3 – Submit
Payload POST https://api.alshumookh-pay.com/api/v1/payloads/ingest Content-Type: application/json
Authorization: Bearer eyJ... X-Timestamp: 2026-05-11T10:00:00Z X-Signature: X-Client-ID: YOUR_CLIENT_ID
Body: { ... payload body ... }
```

5.3 API Response Codes

HTTP	Status	Error Code	Meaning	Action Required
200	accepted	–	Payload received — processing initiated	Await webhook callback
422	error	validation_error	Missing or invalid field in payload	Fix field and resubmit
401	error	invalid_api_key	API Key missing or invalid	Check X-API-Key header
401	error	hmac_required	X-Signature header missing	Add HMAC-SHA256 signature
401	error	invalid_hmac	HMAC signature does not match	Verify signature computation
401	error	oauth_required	OAuth2 Bearer token required	Obtain token via /oauth/token
409	error	duplicate_reference	sender_reference already processed	Use a new unique reference
429	error	rate_limited	Too many requests	Retry after 60 seconds
500	error	server_error	Internal server error	Contact support immediately

5.4 Outbound Signed Webhook — Callback to Sender

```
POST https://your-system.com/webhook/callback Content-Type: application/json X-ALSHUMOOKH-Signature:
X-Timestamp: 2026-05-11T10:05:00Z { "event": "payload.settled", "payload_id": "PLY-XXXX",
"sender_reference": "TXN-2026-001", "status": "settled", "amount": 1000000.00, "currency": "USDT",
"network": "ERC20", "transaction_hash": "0xCONFIRMED_TX_HASH", "block_number": 19500000, "settled_at":
"2026-05-11T10:04:55Z", "distribution": [
{"wallet": "0xWALLET_A", "amount": 400000, "percentage": 40, "tx_hash": "0x..."},
{"wallet": "0xWALLET_B", "amount": 350000, "percentage": 35, "tx_hash": "0x..."},
{"wallet": "0xWALLET_C", "amount": 250000, "percentage": 25, "tx_hash": "0x..."} ] }
```

5.5 Payload Validation Rules

Field / Rule	Condition	Error Code if Violated
sender_reference	Unique per client — previously seen values rejected	duplicate_reference
amount	Positive decimal number greater than 0	invalid_amount
currency	Must be exactly "USDT"	invalid_currency
network	Must be "ERC20" or "TRC20"	invalid_network
beneficiary_wallets	1–5 items. All percentages must sum to exactly 100	invalid_pgl
transaction_hash	Must resolve on-chain for the specified network	unverifiable_hash
timestamp	Must be within ±5 minutes of server UTC time	timestamp_expired
receiver_wallet	Must exactly match ALSHUMOOKH's registered wallet	wallet_mismatch
X-Signature	Required when client flag hmac_required = true	hmac_required



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